Introduction to Deep Learning

Session 8: Classification & Maximum Likelihood

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Lecture Overview

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1. Classification Problem
2. Binary Classification
3. Multiclass Classification
4. Maximum Likelihood
5. Information Theory

From Regression to Classification

Regression: predict continuous value $y \in \mathbb{R}$

- Output: house price, temperature, critical temperature
- Loss: MSE — penalises distance between prediction and target

Classification: predict discrete category $y \in \{1, \ldots, K\}$

- Output: image class, spam or not, medical diagnosis
- Output space fundamentally different — no notion of distance between classes

Problem: raw network output $f_{\boldsymbol{\theta}}(\mathbf{x}) \in \mathbb{R}$ cannot be interpreted as class directly

Idea: convert output to probability distribution over classes

What We Need

Requirements for valid probability distribution over K classes:

$$p_k \geq 0 \quad \forall k \qquad \text{and} \qquad \sum_{k=1}^{K} p_k = 1$$

Raw network outputs (logits) $\mathbf{z} = f_\theta(\mathbf{x}) \in \mathbb{R}^K$

Solution: transformation that maps logits to valid probability distribution

- **Binary classification** ($K = 2$): **sigmoid** → next section
- **Multiclass classification** ($K > 2$): **softmax** → following section

Loss function: for probability outputs? not MSE, hm ...

Binary Classification

Sigmoid

Binary classification: $y \in \{0, 1\}$ - network outputs single logit $z = f_\theta(\mathbf{x}) \in \mathbb{R}$

Sigmoid: squashes logit to $(0, 1)$ - interpreted as probability of class 1

$$\hat{p} = \sigma(z) = \frac{1}{1 + e^{-z}}$$

- $\sigma(z) \in (0, 1)$ for all $z \in \mathbb{R}$
- $\sigma(0) = 0.5$ - symmetric around origin
- $\sigma(z) \to 1$ as $z \to \infty$
- $\sigma(z) \to 0$ as $z \to -\infty$

Binary Cross-Entropy

Loss: penalise confident wrong predictions

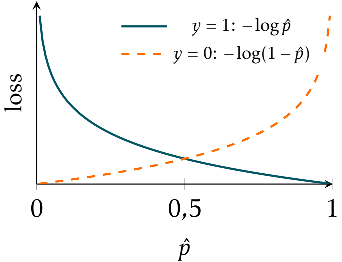
$$\ell(\hat{p}, y) = -y \log \hat{p} - (1-y) \log(1-\hat{p})$$

$y = 1$: loss $= -\log \hat{p}$

- $\hat{p} \to 1$: loss $\to 0$ ✓
- $\hat{p} \to 0$: loss $\to \infty$ ×

$y = 0$: loss $= -\log(1-\hat{p})$

- $\hat{p} \to 0$: loss $\to 0$ ✓
- $\hat{p} \to 1$: loss $\to \infty$ ×



Training loss:

$$\mathcal{L}(\boldsymbol{\theta}) = -\frac{1}{N} \sum_{i=1}^{N} \left[y^{(i)} \log \hat{p}^{(i)} + (1-y^{(i)}) \log(1-\hat{p}^{(i)}) \right]$$

Multiclass Classification

Softmax

Multiclass classification: $y \in \{1, \ldots, K\}$ - network outputs K logits $\mathbf{z} \in \mathbb{R}^K$

Softmax: maps logits to valid probability distribution

$$\hat{p}_k = \frac{e^{z_k}}{\sum_{j=1}^{K} e^{z_j}}, \qquad k = 1, \ldots, K$$

- $\hat{p}_k > 0$ for all k — all probabilities positive
- $\sum_{k=1}^{K} \hat{p}_k = 1$ — valid distribution
- Larger logit $\Rightarrow$ larger probability

Connection to sigmoid: softmax with $K = 2$ reduces to sigmoid

$$\hat{p}_1 = \frac{e^{z_1}}{e^{z_1} + e^{z_2}} = \frac{1}{1 + e^{-(z_1 - z_2)}} = \sigma(z_1 - z_2)$$

Single logit $z = z_1 - z_2$ — one degree of freedom is redundant

Categorical Cross-Entropy

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Labels: one-hot vector $\mathbf{y} \in \{0,1\}^K$ — $y_k = 1$ for correct class, 0 otherwise

Loss: picks out log probability of correct class

$$\ell(\hat{\mathbf{p}}, \mathbf{y}) = -\sum_{k=1}^{K} y_k \log \hat{p}_k = -\log \hat{p}_y$$

- Correct class predicted confidently $\hat{p}_y \to 1$: loss $\to 0$ ✓
- Correct class predicted with low confidence $\hat{p}_y \to 0$: loss $\to \infty$ ×

Training loss:

$$\mathcal{L}(\boldsymbol{\theta}) = -\frac{1}{N} \sum_{i=1}^{N} \log \hat{p}^{(i)}_{y^{(i)}}$$

Maximum Likelihood

Maximum Likelihood Principle

Setup: model defines conditional distribution $p(y \mid \mathbf{x}; \boldsymbol{\theta})$ over outputs

Likelihood: probability of observed data as a **function of parameters** — data $\mathcal{D}$ is fixed

$$L(\boldsymbol{\theta}) = \prod_{i=1}^{N} p(y^{(i)} \mid \mathbf{x}^{(i)}; \boldsymbol{\theta})$$

MLE: find parameters that make observed data most probable

$$\boldsymbol{\theta}^* = \arg\max_{\boldsymbol{\theta}} \; L(\boldsymbol{\theta})$$

Log-likelihood: sum instead of product — same maximiser, more convenient

$$\log L(\boldsymbol{\theta}) = \sum_{i=1}^{N} \log p(y^{(i)} \mid \mathbf{x}^{(i)}; \boldsymbol{\theta})$$

From MLE to Loss Minimisation

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Maximise log-likelihood ≡ **minimise** negative log-likelihood (NLL)

$$\boldsymbol{\theta}^* = \arg\max_{\boldsymbol{\theta}} \log L(\boldsymbol{\theta}) = \arg\max_{\boldsymbol{\theta}} \sum_{i=1}^{N} \log p(y^{(i)} \mid \mathbf{x}^{(i)}; \boldsymbol{\theta}) = \arg\min_{\boldsymbol{\theta}} \underbrace{-\frac{1}{N} \sum_{i=1}^{N} \log p(y^{(i)} \mid \mathbf{x}^{(i)}; \boldsymbol{\theta})}_{\mathcal{L}(\boldsymbol{\theta})}$$

Key insight: choice of $p(y \mid \mathbf{x}; \boldsymbol{\theta})$ determines the loss function

Distributional assumption	**Loss function**
Gaussian	MSE
Bernoulli	Binary cross-entropy
Categorical	Categorical cross-entropy

Gaussian Assumption → MSE

Assume: $p(y \mid \mathbf{x}; \boldsymbol{\theta}) = \mathcal{N}(y; f_{\boldsymbol{\theta}}(\mathbf{x}), \sigma^2) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(y - f_{\boldsymbol{\theta}}(\mathbf{x}))^2}{2\sigma^2}\right)$

Log-likelihood: $\log p(y \mid \mathbf{x}; \boldsymbol{\theta}) = -\frac{(y - f_{\boldsymbol{\theta}}(\mathbf{x}))^2}{2\sigma^2} - \frac{1}{2}\log(2\pi\sigma^2)$

NLL: $-\frac{1}{N}\sum_{i=1}^{N} \log p(y^{(i)} \mid \mathbf{x}^{(i)}; \boldsymbol{\theta}) = \frac{1}{2\sigma^2} \underbrace{\frac{1}{N}\sum_{i=1}^{N}(y^{(i)} - f_{\boldsymbol{\theta}}(\mathbf{x}^{(i)}))^2}_{\text{MSE}} + \text{const}$

Minimising NLL under Gaussian noise assumption ≡ minimising MSE

Connects to Session 3: MSE is not arbitrary — it is MLE under Gaussian noise

Bernoulli Assumption → Binary Cross-Entropy

Assume: $p(y \mid \mathbf{x}; \boldsymbol{\theta}) = \text{Bernoulli}(y; \hat{p}) = \hat{p}^{\,y}(1-\hat{p})^{1-y}, \quad \hat{p} = \sigma(f_{\boldsymbol{\theta}}(\mathbf{x}))$

Log-likelihood: $\log p(y \mid \mathbf{x}; \boldsymbol{\theta}) = y \log \hat{p} + (1-y)\log(1-\hat{p})$

NLL: $-\frac{1}{N}\sum_{i=1}^{N} \log p(y^{(i)} \mid \mathbf{x}^{(i)}; \boldsymbol{\theta}) = \underbrace{-\frac{1}{N}\sum_{i=1}^{N}\left[y^{(i)}\log \hat{p}^{(i)} + (1-y^{(i)})\log(1-\hat{p}^{(i)})\right]}_{\text{binary cross-entropy}}$

Minimising NLL under Bernoulli assumption ≡ minimising binary cross-entropy

Categorical Assumption → Categorical Cross-Entropy

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Assume: $$p(y \mid \mathbf{x}; \boldsymbol{\theta}) = \text{Categorical}(y; \hat{\mathbf{p}}) = \prod_{k=1}^{K} \hat{p}_k^{y_k} = \hat{p}_y, \quad \hat{\mathbf{p}} = \text{softmax}(f_{\boldsymbol{\theta}}(\mathbf{x}))$$

Log-likelihood: $$\log p(y \mid \mathbf{x}; \boldsymbol{\theta}) = \sum_{k=1}^{K} y_k \log \hat{p}_k = \log \hat{p}_y$$

NLL: $$-\frac{1}{N}\sum_{i=1}^{N} \log p(y^{(i)} \mid \mathbf{x}^{(i)}; \boldsymbol{\theta}) = \underbrace{-\frac{1}{N}\sum_{i=1}^{N} \log \hat{p}^{(i)}_{y^{(i)}}}_{\text{categorical cross-entropy}}$$

Minimising NLL under categorical assumption ≡ minimising categorical cross-entropy

Information Theory

Why Information Theory?

So far: cross-entropy - mathematical result of MLE derivation

Open question: what does cross-entropy *mean*?

Answer - information theory:

- Cross-entropy: fundamental measure of how well distribution q approximates distribution p
- Training minimises gap between model distribution $q = \hat{\mathbf{p}}$ and true distribution p

We will build up:

- **Information content** — how surprising is an event?
- **Entropy** — how uncertain is a distribution?
- **Cross-entropy and KL divergence** — how far apart are two distributions?

Payoff: MLE, cross-entropy loss, and KL divergence - three views of same objective

Information Content

Intuition: how surprised are we when event k occurs?

- Rare event — very surprising — highly informative
- Common event — expected — low information

Information content (Shannon, 1948):

$$I(k) = -\log p_k$$

p_k	$I(k)$	Interpretation
1	0	certain event — no surprise
0.5	1 bit	coin flip
0.01	≈ 6.6 bits	rare event — very surprising
$\to 0$	$\to \infty$	impossible event becoming true

Information Content

Intuition: how surprised are we when event k occurs?

- Rare event — very surprising — highly informative
- Common event — expected — low information

Information content: $I(k) = -\log_2 p_k$ - optimal number of bits to encode event k

p_k	$I(k)$	Interpretation
1	0	certain event — no bits needed
0.5	1 bit	coin flip
0.01	≈ 6.6 bits	rare event — very surprising - long encoding
$\to 0$	$\to \infty$	impossible event becoming true - impossible to encode

Why $-\log$? More probable events - shorter codes; info of independent events - sum of info; monotonic and smooth.

Entropy

Entropy: expected information content = minimum average code length for samples from distribution p

$$H(p) = \mathbb{E}_{k\sim p}[I(k)] = -\sum_k p_k \log p_k$$

Measures uncertainty:

- High entropy — high uncertainty, long average code — distribution spread out
- Low entropy — low uncertainty, short average code — distribution concentrated
- $H(p) = 0$ — deterministic distribution, no bits needed
- $H(p)$ maximised by uniform distribution — maximum uncertainty

Binary Entropy and Coding

Binary distribution: $\boldsymbol{p} = (p, 1-p)$

$$H(p) = -p\log_2 p - (1-p)\log_2(1-p)$$

- **Fair coin** $p = 0.5$: $H = 1$ bit — optimal code 1 bit per outcome; matches practice
- **Biased coin** $p = 0.9$: $H \approx 0.47$ bits — in theory we can do better than 1 bit per symbol

Practical limit: single outcomes cannot be encoded in less than 1 bit

The bound $H(p) < 1$ is only achievable asymptotically — by encoding long sequences of outcomes together rather than individual symbols

This motivates a figure: $H(p)$ as a function of $p \in [0,1]$ — symmetric, maximised at $p = 0.5$, zero at $p \in \{0,1\}$

Entropy

Entropy: expected information content of distribution p

$$H(p) = \mathbb{E}_{k\sim p}[I(k)] = -\sum_k p_k \log p_k$$

Measures uncertainty:

- High entropy — high uncertainty, distribution spread out
- Low entropy — low uncertainty, distribution concentrated
- $H(p) = 0$ — deterministic distribution, no uncertainty
- $H(p)$ maximised by uniform distribution

Example: binary distribution $p = (p, 1 - p)$

$$H(p) = -p \log p - (1 - p) \log(1 - p)$$

Maximised at $p = 0.5$ — complete uncertainty between two outcomes

Cross-Entropy

Setup: true distribution p, model distribution q

Cross-entropy: expected information content when using q to encode events from p

$$H(p,q) = -\sum_k p_k \log q_k$$

- $H(p,q) \geq H(p)$ — using wrong distribution always costs extra bits
- $H(p,q) = H(p)$ iff $p = q$ — no extra cost when model is perfect

Connection to classification loss:

- True labels $\Rightarrow$ true distribution p (one-hot)
- Model predictions $\Rightarrow$ estimated distribution $q = \hat{\mathbf{p}}$
- Categorical cross-entropy loss $= H(p,q)$

KL Divergence

KL divergence: extra cost of using q instead of p

$$D_{KL}(p\|q) = H(p,q) - H(p) = \sum_k p_k \log \frac{p_k}{q_k}$$

- $D_{KL}(p\|q) \geq 0$ — always non-negative
- $D_{KL}(p\|q) = 0$ iff $p = q$
- Not symmetric: $D_{KL}(p\|q) \neq D_{KL}(q\|p)$ in general

Connection to cross-entropy loss: since $H(p)$ is fixed (labels do not depend on $\boldsymbol{\theta}$)

$$\min_{\boldsymbol{\theta}} H(p,q) \equiv \min_{\boldsymbol{\theta}} D_{KL}(p\|q)$$

Minimising cross-entropy loss ≡ minimising KL divergence between true and predicted distribution

Three Views, One Objective

Three perspectives on the same optimisation problem:

Perspective	Objective
Maximum likelihood	maximise probability of observed data
Information theory	minimise KL divergence $D_{KL}(p \| q)$
Loss minimisation	minimise cross-entropy $H(p, q)$

All three are equivalent — MLE gives the statistical motivation,
cross-entropy gives the practical loss, KL divergence gives the geometric interpretation

The choice of loss function is not arbitrary — it reflects an assumption about the data-generating process